

An analytic model of epistemic collapse

Helivan Calcifer Computing

working note

Summary. Because agents relay verbatim and an LLM’s answering behavior is a measured multinomial over a small set of context states, the probability of epistemic collapse can be computed rather than simulated. We derive (i) the probability that a single agent can answer a question, from a renewal argument on its memory; (ii) the probability that a question is in collapse, from a birth–death chain over how many agents hold it; and (iii) the probability of χ -epistemic collapse, from a binomial count over questions. The model has no fitted parameters and reproduces simulation for three strategies — no mitigation, sharing secondhand information, and books — across two decades of environment size (Fig. 1). To leading order it yields a closed-form law for the *epistemic capacity* $M^*(\chi)$ — the largest environment a system can sustain with less than a proportion χ of its questions in collapse — and for the *epistemic efficiency* of each mitigation, $\Gamma(\chi) = M_{\text{strategy}}^*(\chi)/M_{\text{none}}^*(\chi)$: sharing secondhand information raises capacity by a factor $x/(1 - e^{-x})$ with $x = K \ln(1/\chi)/(NE)$ — about 10–20% here — while books raise it by $\approx (1 - p_w)(1 + fKb/E)$ — about 3.5 $\times$ here (Sec. 4).

1 Model and notation

N agents; an environment of M questions; each agent spends B actions per step, of which a fraction α query the environment ($E = \alpha B$ per step, questions drawn uniformly) and $K = (1 - \alpha)B$ ask peers. Asks target questions the asker cannot currently answer. Memory is unbounded; retrieval keeps the k entries with the largest $\langle q, k \rangle e^{-c \Delta t}$. An exact-match entry scores 1 at insertion, any other entry scores at most $\sigma < 1$, so an entry received a steps ago is outscored by a newer entry only after $\Delta = \ln(1/\sigma)/c$ steps. Write $D = \lfloor \Delta \rfloor$.

Retention. An entry for q inserted a steps ago is retrievable iff $a \leq D$, or fewer than k new entries have been inserted in the $a - D$ steps since. Each of the B actions inserts an entry with probability q_{ins} (an observation always inserts; an ask inserts iff it is answered), so the count over m steps is $\text{Bin}(mB, q_{\text{ins}})$. Forgetting is therefore *activity-dependent*: the busier the agent, the faster old entries are crowded out.

Gate. Given its context, an agent answers or abstains. For the LLM we measured $P(\text{IDK} \mid q \text{ in context}) = \varepsilon$ (≈ 0.03 – 0.05) and $P(\text{answer} \mid q \text{ not in context}) \approx 0$; the toy has $\varepsilon = 0$.

2 Results

Proposition 1 (Single agent). *Let $\mu = 1 - (1 - \alpha/M)^B$ be the per-step probability that an agent observes q . With observation as the only way to (re)acquire q , the stationary probability that the agent cannot answer q is*

$$p = \sum_{a>D} \mu(1 - \mu)^a \Pr[\text{Bin}((a - D)B, q_{\text{ins}}) \geq k], \quad q_{\text{ins}} = \alpha + (1 - \alpha)(1 - p), \quad (1)$$

solved as a fixed point (asks are answered when the peer holds the question).

The age of the agent’s last acquisition is geometric; the sum integrates the retention rule over that age.

Proposition 2 (Question-level collapse). *Let n be the number of agents that can answer q . Holders lose q at rate $\delta = \mu p/(1 - p)$ each (the exponential rate reproducing Eq. 1); each non-holder acquires it at rate $g(n)$. The stationary law of the birth–death chain is*

$$\pi_n \propto \prod_{j=1}^n \frac{(N - j + 1) g(j - 1)}{j \delta}, \quad (2)$$

and the question is in epistemic collapse with probability $\pi_0^\varepsilon = \sum_n \pi_n \varepsilon^n$ (every holder also abstains), which is π_0 for a deterministic gate.

The strategies differ only in the acquisition rate:

- **No mitigation.** $g(n) = \mu$: agents are independent, $\pi_0 = p^N$.
- **Sharing secondhand information.** $g(n) = \mu + \frac{K}{Mp} \frac{n}{N-1}$: an agent spends its K asks on its $\approx Mp$ unknown questions and succeeds iff the random peer holds q . Answered asks also insert, so $q_{\text{ins}} = \alpha + (1-\alpha) \Pr[\text{peer holds } q \mid \text{asker lacks } q]$ from the chain.
- **Books.** $g(n) = \mu' + \frac{fK'}{Mp} b$, where a fraction f of asks read the book, b is the fraction of questions the book covers, and a write tax p_w scales the budget ($\mu', K' \leftarrow (1-p_w)\mu, (1-p_w)K$). The book never forgets; it gains an uncovered question when a committing writer holds it, and commits arrive at rate Np_w/W , so

$$\frac{d(1-b)}{dt} = -\frac{Np_w}{W} (1-p_u(b)) (1-b), \quad (3)$$

with $p_u(b)$ from Eq. 1 with the reads' extra insertions. Agents are again independent given b .

Proposition 3 (System-level collapse). *Treating questions as independent, the number in collapse is $\text{Bin}(M, \pi_0)$ and*

$$\Pr[\chi\text{-epistemic collapse}] = \Pr[\text{Bin}(M, \pi_0) \geq \lceil \chi M \rceil]. \quad (4)$$

Since the binomial concentrates, this is nearly a step function at $\pi_0 = \chi$: χ -collapse is a sharp transition in the environment size.

Corollary 1 (Epistemic capacity, no mitigation). *For $M \gg B$, $1-p \approx E\tau/M$ with $\tau \approx D+1+k/(Bq_{\text{ins}})$ the mean retention time, so $\pi_0 = p^N \approx e^{-NE\tau/M}$ and the system enters χ -collapse at*

$$M^* \approx \frac{NE\tau}{\ln(1/\chi)}. \quad (5)$$

The system survives while its *collective observational memory* — agents $\times$ observations per step $\times$ steps each observation is retained — exceeds the environment size by a factor $\ln(1/\chi)$. Memory decay enters only through τ ; environment speed does not enter at all, which is why speed moves staleness but not collapse.

3 Validation

Figure 1 compares the model with the simulator ($N = 10$, $B = 11$, $E = 3$, $k = 3$, $\sigma = 0.5$, 16 seeds, steady state over the final 200 of 600 steps). No parameter is fitted. Across all three strategies, both decay rates and nine environment sizes, the per-question collapse probability matches to within 0.03, and the location of the χ -collapse transition matches; the one visible miss is mimesis at $c = 0.5$, $M = 50$, which sits on the steep part of the transition. The book coverage $b(t)$ is predicted to within 0.02. Table 1 compares the closed-form M^* with the exact solution; it is within 4% for slow decay and overestimates by $\approx 25\%$ for fast decay, where retention lasts only two to three steps and the discreteness of time matters.

c	χ	M^* exact	$NE\tau/\ln(1/\chi)$
0.05	0.25	307	319
0.05	0.50	618	643
0.5	0.25	47	59
0.5	0.50	98	123

Table 1: Epistemic capacity: exact solution of the model vs. the closed form.

4 How much do mitigations prevent collapse?

In the collapse regime (holders of a question are rare, $M \gg B$) every strategy acts in the same way: it multiplies the system's collective observational memory $NE\tau$ by an amplification factor $\Gamma \geq 1$ (or < 1 if it hurts). Consequently

$$\pi_0^{\text{strategy}} \approx (\pi_0^{\text{none}})^\Gamma, \quad M_{\text{strategy}}^* \approx \Gamma M_{\text{none}}^*, \quad (6)$$

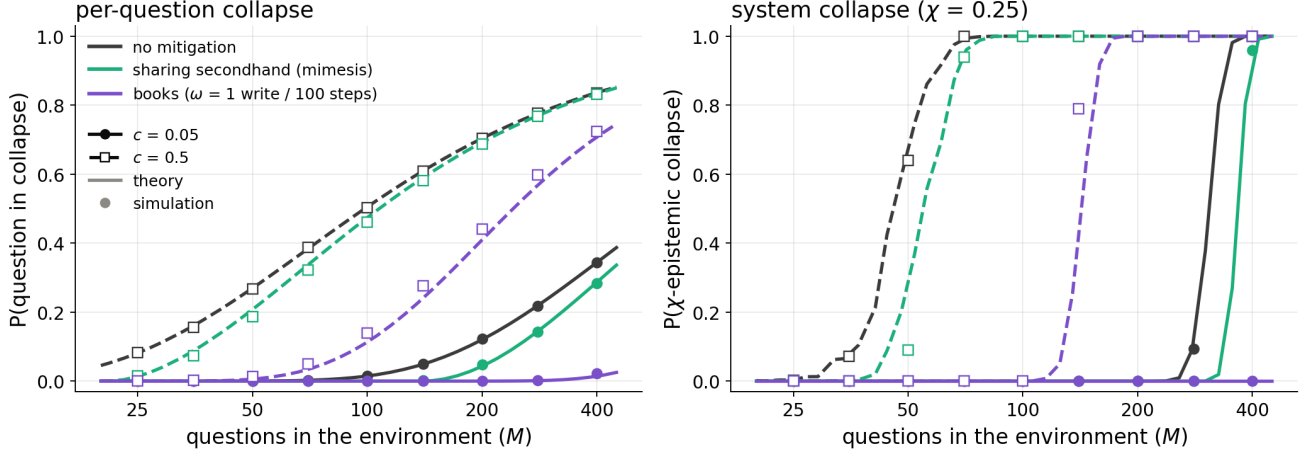


Figure 1: Theory (lines) vs. simulation (markers), no fitted parameters. Left: probability that a question is in epistemic collapse. Right: probability of χ -epistemic collapse, $\chi = 0.25$. Books: one completed write per agent per 100 steps ($p_w = 0.05$, $W = 5$), all asks read the book.

so Γ is the *epistemic efficiency*: the multiple by which a strategy enlarges the environment the system can track before collapsing.

Proposition 4 (Sharing secondhand information). *With per-holder transmission rate $a \approx K/M$ (asks from the $N - 1$ non-holders, answered by one holder), loss rate $1/\tau$ and immigration $N\mu \approx NE/M$ from observation, the number of holders is a branching process with immigration and reproduction number $R = K\tau/M$. Its stationary law is negative binomial, with*

$$\pi_0^{mim} = (1 - R)^{NE\tau/(MR)} = \exp\left(-\frac{NE\tau}{M} \varphi(R)\right), \quad \varphi(R) = \frac{-\ln(1 - R)}{R}, \quad (7)$$

so $\Gamma_{mim} = \varphi(R) \cdot \tau_{mim}/\tau$. At the collapse threshold R is itself fixed by M^* , and solving gives a law that does not involve memory decay at all:

$$\Gamma_{mim}^* = \frac{x}{1 - e^{-x}}, \quad x = \frac{K \ln(1/\chi)}{NE}. \quad (8)$$

Mimesis is therefore strong only when $x \gtrsim 1$: when an agent's ask bandwidth K is large compared with the population's total observation rate NE . It can only amplify evidence that someone has observed, and near the threshold a question is observed so rarely that its few holders forget it before peers ask. In our setting $x = 0.37$ ($\chi = 0.25$) and 0.18 ($\chi = 0.5$): mimesis enlarges the trackable environment by only 20% and 10%. The benefit shrinks as the population or its observation rate grows, and as the collapse proportion becomes stricter.

Proposition 5 (Books). *Agents are independent given book coverage $b(t)$, and a read of a covered question acquires it exactly as an observation does. Hence books add $fK'b$ to the per-agent source rate, at the cost of the write tax and of extra crowding:*

$$\Gamma_{books}(t) = (1 - p_w) \frac{\tau_b}{\tau} \left(1 + \frac{fK b(t)}{E}\right) \xrightarrow{t \rightarrow \infty} (1 - p_w) \frac{\tau_b}{\tau} \frac{E + fK}{E}, \quad (9)$$

with $b(t)$ from the coverage equation and τ_b the retention time under the heavier insertion load of reading.

A book turns the peer-ask budget into source bandwidth: with every ask reading the book ($f = 1$) the steady-state factor is $\approx (1 - p_w) \tau_b/(\tau \alpha)$ — the inverse of the fraction of the budget spent observing. In our setting $\Gamma_{books} \approx 3.5$. The write rate $\omega = p_w/W$ controls only how fast Γ is reached, through $b(t)$; the write probability p_w additionally taxes the steady state.

The closed forms are rare-holder limits: they are most accurate near the collapse threshold, which is where they are needed, and conservative (they slightly underestimate efficiency) when agents hold a substantial share of the environment. Against simulation, π_0^{none} raised to Γ predicts the collapsed fraction under mimesis to within 0.03 and under books to within 0.03 once the crowding ratio τ_b/τ is included.

strategy	c	$\chi = 0.25$		$\chi = 0.5$	
		exact	closed form	exact	closed form
mimesis	0.05	1.19	1.23	1.09	1.10
mimesis	0.5	1.19	1.22	1.09	1.10
books ($b = 1$)	0.05	3.62	3.48	3.51	3.48
books ($b = 1$)	0.5	3.39	3.48	3.15	3.48

Table 2: Epistemic efficiency $\Gamma = M_{\text{strategy}}^*/M_{\text{none}}^*$: exact solution of the chain model vs. closed form. Books: $p_w = 0.05$, $f = 1$, full coverage.

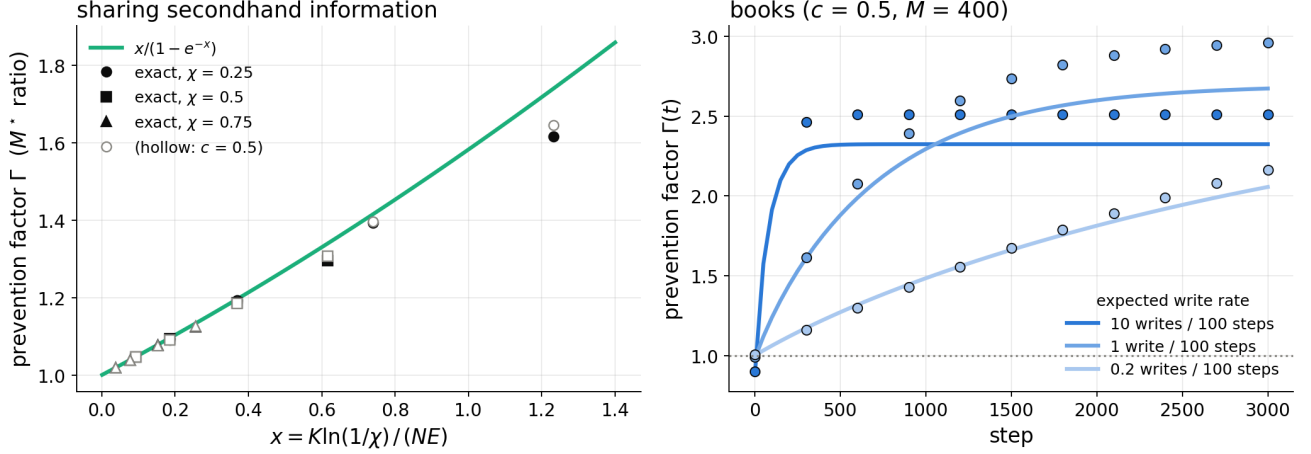


Figure 2: Epistemic efficiency. Left: mimesis at the collapse threshold collapses onto the one-parameter curve $x/(1 - e^{-x})$ across population sizes, decay rates and collapse proportions (markers: exact model). Right: books over time for three write rates at $c = 0.5$, $M = 400$; the book's value accumulates as it fills in (lines: closed form; markers: exact model). The closed form is exact while the book fills and underestimates the plateau by ≈ 8 –10% once coverage is complete, when holders are no longer rare.

Caveats. (1) Questions are treated as independent; they are coupled weakly through shared insertion rates. (2) Losses are modeled at a constant rate chosen to reproduce the exact single-agent occupancy; the true loss hazard has a safe period of D steps. (3) Asks are assumed spread over unknown questions and capped by nothing; this overpredicts mimesis at small M (agent-level abstention), not collapse. (4) The book result assumes that reading the book re-grounds a forgotten fact even when the agent once held newer, now crowded-out evidence (`book_reground=True`); the simulator's default does not, which the model reproduces only with that flag.